

# Churn Prediction and Retention Strategy for a Telecom Provider

## Project Description:

**Company:** ConnectTel Communications Ltd

**Industry:** telecommunications - Mobile & Broadband Services

ConnectTel is a pan-India telecom operator offering mobile, broadband, and bundled data plans to both prepaid and postpaid customers. Over the last 12–18 months, the company has seen increasing customer churn, especially in high-value segments such as heavy data users and long-tenure postpaid customers.

Despite aggressive discounts, bundled OTT partnerships, and periodic promotional campaigns, the leadership team has identified three worrying trends:

- A steady rise in monthly churn rate, particularly among price-sensitive and digitally savvy customers
- High-value customers silently migrating to competitors without raising complaints
- Reactive retention — the business tries to win customers back only *after* they initiate port-out or disconnect requests

The Chief Commercial Officer (CCO) has mandated a data-driven churn prediction and retention strategy. The objective is to:

- Predict which customers are at high risk of churn in the near future
- Understand why they are likely to churn – e.g., poor service experience, billing shocks, price dissatisfaction, low engagement
- Design targeted offers and interventions (retention actions) to reduce churn and protect customer lifetime value (CLV)

The project will use customer-level data such as demographics, plan details, usage patterns (voice, data, SMS), payment behaviour, contract type, complaint and service interaction history, and tenure with the operator.

Learners will build classification models to estimate churn probability and translate these into actionable retention strategies.

Stakeholders include:

- **Customer Experience & Retention Team**  
Wants early visibility of “at risk” customers to design timely outbound calls, personalized offers, and proactive save campaigns.
- **Marketing & Product Team**  
Needs to understand which segments are churning and why, and to refine tariff plans, bundles, and loyalty programs accordingly.
- **Network Operations & Quality of Service (QoS) Team**  
Needs insights into whether network quality (dropped calls, low data speeds, outages) is a major churn driver in specific circles/regions.
- **Sales & Channel Partners**  
Want to identify high-risk customers in their base to prioritize engagement and upsell/cross-sell before competitors step in.
- **Finance & Strategy Team**  
Focused on defending revenue, optimizing acquisition vs retention spend, and improving overall customer lifetime value.

A strong solution should help these teams **reduce avoidable churn**, improve retention efficiency, and sharpen focus on the most valuable customers to save.

A successful project delivers tangible uplift in:

- Early identification of high-risk customers (higher recall for churners)
- Reduction in monthly churn rate across key segments (prepaid, postpaid, high-value customers)
- Improved retention efficiency by targeting only customers who are truly at risk
- Higher customer lifetime value (CLV) through proactive engagement
- Operational efficiency for retention teams by allocating resources to the right customers at the right time

ConnectTel needs a predictive churn detection model capable of distinguishing customers likely to churn from those who will stay—based on patterns in customer tenure, plan details, usage behaviour, payment consistency, complaint history, service experience, and engagement indicators.

The challenge is heightened because:

- Churn behaviour is multi-factorial and varies across segments
- Some customers leave without expressing dissatisfaction (silent churners)
- Usage patterns and billing behaviour can resemble both loyal and churn-prone customers
- Churn is typically imbalanced, with only a small percentage leaving each cycle
- Drivers of churn differ across prepaid/postpaid, regions, age groups, and customer value tiers

Your task is to design a robust ML-based churn classification system that can be embedded into a proactive retention workflow.

**Learners must validate the dataset, perform exploratory churn analysis, engineer meaningful behavioural and service-related features, and build a machine learning model to predict customer churn. They must evaluate model performance, interpret key churn drivers, and present clear, segment-wise retention recommendations. A concise business report summarizing insights, model rationale, and proposed customer retention actions is required.**

## Data Definition:

Use the file **[Data Definition.xlsx]** to understand fields, key relationships, and business rules.

## Instruction to Learners:

### Validate Data Integrity

Check all datasets for:

- Primary & foreign key consistency
- Null or incorrect values
- Logical/business rule violations
- Date correctness and chronological order. Make corrections wherever required.

### Perform Feature Engineering

Create additional features needed for churn analysis and classification (e.g., usage drop indicators, payment delay metrics, complaint frequency, plan/value segmentation, tenure buckets, bill shock indicators, and engagement scores).

# Suggested Feature Engineering Guide (For Reference Only)

*This section provides business context behind a set of commonly used engineered features in telecom churn prediction. The guide is illustrative and helps frame how different behavioural, billing, usage, and service-experience variables can influence a customer's likelihood to churn.*

## 1. Usage Behaviour Features

### Usage Drop Indicators

Customers who suddenly reduce their data or voice usage often signal an intention to leave or shift activity to another operator.

### High Data Consumption Flag

Heavy users are high-value customers; when they churn, the financial impact is significant. Their behaviour patterns often differ from low-usage customers.

### Peak-Hour Usage Patterns

Declining usage during peak hours may indicate dissatisfaction with network speed or congestion.

## 2. Billing & Payment Features

### Bill Shock Indicator

A sudden increase in monthly charges (e.g., exceeding data limits) can trigger churn—especially in postpaid users.

### Payment Delay Metrics

Consistently late bill payments or increasing delays may indicate disengagement or financial stress.

## **Outstanding Balance or Unpaid Dues**

Customers with accumulating dues are statistically more likely to churn or switch providers.

## **3. Customer Tenure & Lifecycle Features**

### **Tenure Bucket**

Short-tenure customers tend to be more price-sensitive and churn at higher rates, while long-tenure customers churn mainly due to service issues.

### **Loyalty Tier or Reward Eligibility**

Customers in lower tiers may feel undervalued, increasing churn risk.

## **4. Service Quality & Complaint Features**

### **Complaint Frequency**

Frequent complaints about network, billing errors, or customer service strongly correlate with churn.

### **Issue Resolution Time**

Slow resolution erodes trust and increases churn propensity.

### **Network Quality Indicators**

Dropped calls, low data speeds, and outages are common triggers for switching operators.

## **5. Product & Plan Features**

### **Plan Type (Prepaid/Postpaid)**

Churn behaviour differs sharply by segment—postpaid churn is more value-driven; prepaid churn is more price-driven.

### **Plan Fit Score**

Measures how well the user's plan aligns with their actual usage; mismatch increases churn likelihood.

### **Competitor-Attractive Categories**

Unlimited voice/data users may be targeted more aggressively by competing operators.

## **6. Customer Engagement Features**

### **App/Portal Login Frequency**

Low engagement often signals lower attachment to the operator and higher churn risk.

### **Response to Marketing Offers**

Customers who rarely redeem offers or ignore campaigns may be harder to retain.

### **Service Interaction Depth**

High interaction without problem resolution is a major churn signal.

## **7. Financial Value Features**

### **ARPU (Average Revenue Per User)**

High-value customers must be prioritized for retention; their churn drivers differ from low-value segments.

### **Customer Lifetime Value (CLV) Proxy**

Helps decide which churn-risk customers deserve proactive retention intervention.